

Mixture-Only 17-Class Comparison

Original real mixtures + PT2 real mixtures

Dataset. This comparison uses mixture data only: the 580-sample original real-mixture dataset plus the 324-sample PT2 real-mixture dataset, for 904 total mixture spectra. All scoring is still done in the full 17-class PT2-expanded inference space.

Selection rule. “Best clean classical” is chosen from the non-hand-tuned classical baselines on this 904-sample mixture-only set. “Best clean deep” is chosen from the non-hand-tuned pure-deep baselines on this same set. The localized classical fallback variants and the hybrid deep-plus-classical reranker are intentionally excluded.

Table 1: Full 17-class mixture-only comparison across original and PT2 real mixtures.

Model	Exact match	Micro-precision	Micro-recall	Micro-F1	Avg. returned cher
Siamese+MLP 2-head (default 0.5)	0.306	0.614	0.887	0.726	2.889
Siamese+MLP 2-head (calibrated)	0.301	0.572	0.905	0.701	3.166
Best clean classical	0.940	0.970	0.970	0.970	2.000
Best clean deep	0.969	0.983	0.983	0.983	2.000

Table 2: Per-class precision, recall, and F1 on the combined mixture-only dataset. Classes with zero support across both mixture datasets are omitted.

Chemical	Support	Siamese 0.5			Siamese calibrated			Best clean classical			Best clean deep		
		P	R	F1	P	R	F1	P	R	F1	P	R	F1
1-dodecanethiol	243	0.670	0.893	0.765	0.682	0.979	0.804	1.000	0.967	0.983	1.000	0.942	0.970
6-mercapto-1-hexanol	144	0.330	0.250	0.285	0.295	0.972	0.452	1.000	0.750	0.857	1.000	1.000	1.000
acetonitrile	72	0.667	1.000	0.800	0.667	1.000	0.800	1.000	1.000	1.000	1.000	1.000	1.000
benzene	301	1.000	1.000	1.000	1.000	0.880	0.936	1.000	1.000	1.000	1.000	1.000	1.000
benzenethiol	72	1.000	1.000	1.000	1.000	0.958	0.979	0.667	1.000	0.800	1.000	1.000	1.000
dichloromethane	108	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
diethylamine	36	0.279	0.944	0.430	0.279	0.944	0.430	0.800	1.000	0.889	0.706	1.000	0.828
etoh	121	0.606	0.992	0.752	0.723	0.818	0.767	1.000	1.000	1.000	1.000	1.000	1.000
meoh	243	1.000	0.872	0.932	1.000	0.901	0.948	1.000	0.959	0.979	1.000	0.934	0.966
n,n-dimethylformamide	72	0.889	1.000	0.941	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
n-hexane	108	0.727	0.667	0.696	0.727	0.667	0.696	1.000	1.000	1.000	1.000	1.000	1.000
pyridine	252	1.000	1.000	1.000	1.000	0.845	0.916	1.000	1.000	1.000	1.000	1.000	1.000
toluene	36	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000

Selected clean classical model. pair_nnl replicate_dictionary. Candidate scores on the same mixture-only set: pair_nnl replicate_dictionary: exact=0.940, micro-F1=0.970, pair_nnl with_baseline_atoms: exact=0.928, micro-F1=0.964, exhaustive_pair_nnl: exact=0.913, micro-F1=0.956, cardinality_adaptive_nnl: exact=0.896, micro-F1=0.971, full_library_sparse_support_selection: exact=0.753, micro-F1=0.919.

Selected clean deep model. deep_similarity_supervision. Candidate scores on the same mixture-only set: deep_similarity_supervision: exact=0.969, micro-F1=0.983, deep_binary_coefficient_regressor: exact=0.968, micro-F1=0.983, deep_cnn_encoder: exact=0.923, micro-F1=0.960, deep_replicate_decoder: exact=0.905, micro-F1=0.951.